

Franka Emika Robot Control Comparison Report

Automatic Analysis

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1 Overview

This report presents a comprehensive comparison of robot control performance across different:

- Controllers: FBL (Feedback Linearization) and PBC (Passivity-Based Control)
- Uncertainty levels: high, extreme
- Initial conditions: matched vs mismatched
- Gain tuning: optimized vs unoptimized
- Trajectory types: quintic, bang-bang, bang-coast-bang

Total simulations analyzed: 96

2 Controller Performance Summary

2.1 FBL Controller

Metric	Value
Number of simulations	48
Mean EE max error [mm]	34.67
Std EE max error [mm]	21.34
Best EE max error [mm]	10.15
Worst EE max error [mm]	83.23
Mean joint max error [deg]	13.63
Best joint max error [deg]	7.42
Worst joint max error [deg]	20.59

Table 1: Summary statistics for FBL controller

2.2 PBC Controller

Metric	Value
Number of simulations	48
Mean EE max error [mm]	14.12
Std EE max error [mm]	7.33
Best EE max error [mm]	2.40
Worst EE max error [mm]	20.84
Mean joint max error [deg]	5.71
Best joint max error [deg]	0.82
Worst joint max error [deg]	10.00

Table 2: Summary statistics for PBC controller

3 Detailed Performance Tables

Performance tables show FBL and PBC results on consecutive rows, with the better value (lower error) highlighted in **Green**. The layout is optimized for head-to-head comparison.

3.1 Uncertainty Level: HIGH

3.1.1 Trajectory: Quintic

Table 3: Performance comparison for Quintic trajectory with high uncertainty

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-A	10.19	8.53	8.35	2.25
PBC		6.10	5.17	0.89	0.36
<i>Winner: PBC (EE Diff: 4.09 mm)</i>					
FBL	Match/Opt-B	14.54	9.81	8.40	2.44
PBC		6.36	4.57	1.55	0.46
<i>Winner: PBC (EE Diff: 8.18 mm)</i>					
FBL	Match/Unopt-A	20.93	15.93	17.96	5.35
PBC		7.31	6.32	1.06	0.45
<i>Winner: PBC (EE Diff: 13.62 mm)</i>					
FBL	Match/Unopt-B	48.28	28.65	17.47	5.15
PBC		7.87	5.72	1.87	0.57
<i>Winner: PBC (EE Diff: 40.41 mm)</i>					
FBL	Mismatch/Opt-A	20.84	9.90	10.00	2.52
PBC		20.84	9.65	10.00	2.18
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Opt-B	20.84	10.66	10.00	2.63
PBC		20.84	8.11	10.00	1.81
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Unopt-A	20.99	16.31	17.94	5.34
PBC		20.84	9.31	10.00	1.84
<i>Winner: PBC (EE Diff: 0.15 mm)</i>					
FBL	Mismatch/Unopt-B	48.26	29.22	17.57	5.20
PBC		20.84	8.00	10.00	1.53
<i>Winner: PBC (EE Diff: 27.42 mm)</i>					

3.1.2 Trajectory: BB

Table 4: Performance comparison for BB trajectory with high uncertainty

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-A	10.17	8.60	8.08	2.10
PBC		6.10	5.25	0.89	0.36
<i>Winner: PBC (EE Diff: 4.07 mm)</i>					
FBL	Match/Opt-B	14.04	9.72	8.34	2.40
PBC		6.38	4.54	1.55	0.46
<i>Winner: PBC (EE Diff: 7.66 mm)</i>					
FBL	Match/Unopt-A	21.04	16.05	17.88	5.30
PBC		7.31	6.29	1.06	0.44
<i>Winner: PBC (EE Diff: 13.74 mm)</i>					
FBL	Match/Unopt-B	48.51	28.45	17.44	5.10
PBC		7.89	5.73	1.86	0.57
<i>Winner: PBC (EE Diff: 40.61 mm)</i>					
FBL	Mismatch/Opt-A	20.84	10.06	10.00	2.44
PBC		20.84	9.71	10.00	2.17
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Opt-B	20.84	10.74	10.00	2.66
PBC		20.84	7.80	10.00	1.72
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Unopt-A	20.84	16.25	17.68	5.29
PBC		20.84	9.89	10.00	2.01
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Unopt-B	48.51	28.88	17.54	5.16
PBC		20.84	8.12	10.00	1.55
<i>Winner: PBC (EE Diff: 27.67 mm)</i>					

3.1.3 Trajectory: BCB

Table 5: Performance comparison for BCB trajectory with high uncertainty

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-A	10.15	8.61	8.04	2.10

Table 5 – continued

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
PBC		6.09	5.25	0.89	0.36
<i>Winner: PBC (EE Diff: 4.06 mm)</i>					
FBL	Match/Opt-B	13.68	9.66	8.31	2.36
PBC		6.38	4.47	1.55	0.45
<i>Winner: PBC (EE Diff: 7.31 mm)</i>					
FBL	Match/Unopt-A	21.02	16.01	17.83	5.32
PBC		7.30	6.28	1.06	0.44
<i>Winner: PBC (EE Diff: 13.72 mm)</i>					
FBL	Match/Unopt-B	48.56	28.18	17.42	5.07
PBC		7.85	5.68	1.86	0.57
<i>Winner: PBC (EE Diff: 40.71 mm)</i>					
FBL	Mismatch/Opt-A	20.84	10.05	10.00	2.43
PBC		20.84	9.68	10.00	2.16
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Opt-B	20.84	10.72	10.00	2.63
PBC		20.84	7.86	10.00	1.75
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Unopt-A	20.84	16.25	17.61	5.31
PBC		20.84	9.90	10.00	2.00
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Unopt-B	48.56	28.58	17.54	5.16
PBC		20.84	7.93	10.00	1.48
<i>Winner: PBC (EE Diff: 27.72 mm)</i>					

3.2 Uncertainty Level: EXTREME

3.2.1 Trajectory: Quintic

Table 6: Performance comparison for Quintic trajectory with extreme uncertainty

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-A	19.10	16.21	8.73	3.00
PBC		2.41	1.99	0.84	0.40
<i>Winner: PBC (EE Diff: 16.69 mm)</i>					

Table 6 – continued

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-B	37.83	21.71	7.49	3.05
PBC		11.63	7.35	1.90	0.59
<i>Winner: PBC (EE Diff: 26.20 mm)</i>					
FBL	Match/Unopt-A	39.80	34.00	20.57	6.80
PBC		2.89	2.44	1.03	0.49
<i>Winner: PBC (EE Diff: 36.92 mm)</i>					
FBL	Match/Unopt-B	81.15	51.58	17.27	6.69
PBC		14.56	8.87	2.29	0.72
<i>Winner: PBC (EE Diff: 66.59 mm)</i>					
FBL	Mismatch/Opt-A	20.84	16.62	10.00	3.10
PBC		20.84	8.45	10.00	2.15
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Opt-B	37.83	21.84	10.00	3.17
PBC		20.84	10.10	10.00	1.93
<i>Winner: PBC (EE Diff: 16.98 mm)</i>					
FBL	Mismatch/Unopt-A	39.52	33.95	20.59	6.78
PBC		20.84	7.24	10.00	1.87
<i>Winner: PBC (EE Diff: 18.67 mm)</i>					
FBL	Mismatch/Unopt-B	82.13	48.51	17.28	6.63
PBC		20.84	10.54	10.00	1.67
<i>Winner: PBC (EE Diff: 61.28 mm)</i>					

3.2.2 Trajectory: BB

Table 7: Performance comparison for BB trajectory with extreme uncertainty

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-A	19.05	16.00	7.43	2.77
PBC		2.41	1.99	0.82	0.40
<i>Winner: PBC (EE Diff: 16.64 mm)</i>					
FBL	Match/Opt-B	31.87	20.80	7.47	2.95
PBC		11.67	7.43	1.90	0.59
<i>Winner: PBC (EE Diff: 20.20 mm)</i>					

Table 7 – continued

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Unopt-A	39.27	34.00	20.52	6.79
PBC		2.88	2.40	1.00	0.48
<i>Winner: PBC (EE Diff: 36.39 mm)</i>					
FBL	Match/Unopt-B	83.23	50.97	17.23	6.65
PBC		14.60	8.96	2.28	0.72
<i>Winner: PBC (EE Diff: 68.63 mm)</i>					
FBL	Mismatch/Opt-A	20.84	16.55	10.00	2.94
PBC		20.84	8.24	10.00	2.10
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Opt-B	31.87	20.98	10.00	3.06
PBC		20.84	10.00	10.00	1.86
<i>Winner: PBC (EE Diff: 11.02 mm)</i>					
FBL	Mismatch/Unopt-A	39.07	33.85	20.58	6.76
PBC		20.84	7.98	10.00	2.03
<i>Winner: PBC (EE Diff: 18.22 mm)</i>					
FBL	Mismatch/Unopt-B	81.44	48.80	17.20	6.61
PBC		20.84	10.82	10.00	1.74
<i>Winner: PBC (EE Diff: 60.60 mm)</i>					

3.2.3 Trajectory: BCB

Table 8: Performance comparison for BCB trajectory with extreme uncertainty

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Opt-A	19.02	16.01	7.42	2.77
PBC		2.40	1.99	0.82	0.40
<i>Winner: PBC (EE Diff: 16.62 mm)</i>					
FBL	Match/Opt-B	31.64	20.39	7.46	2.89
PBC		11.74	7.67	1.90	0.59
<i>Winner: PBC (EE Diff: 19.90 mm)</i>					
FBL	Match/Unopt-A	39.23	33.91	20.49	6.74
PBC		2.88	2.40	1.00	0.48
<i>Winner: PBC (EE Diff: 36.35 mm)</i>					

Table 8 – continued

Controller	Config	EE Max [mm]	EE RMS [mm]	Jt Max [deg]	Jt RMS [deg]
FBL	Match/Unopt-B	82.70	50.03	17.31	6.55
PBC		14.55	9.21	2.28	0.71
<i>Winner: PBC (EE Diff: 68.15 mm)</i>					
FBL	Mismatch/Opt-A	20.84	16.56	10.00	2.93
PBC		20.84	8.32	10.00	2.12
<i>Winner: FBL (EE Diff: 0.00 mm)</i>					
FBL	Mismatch/Opt-B	31.64	20.62	10.00	3.02
PBC		20.84	10.29	10.00	1.90
<i>Winner: PBC (EE Diff: 10.80 mm)</i>					
FBL	Mismatch/Unopt-A	39.05	33.90	20.57	6.71
PBC		20.84	7.96	10.00	2.03
<i>Winner: PBC (EE Diff: 18.21 mm)</i>					
FBL	Mismatch/Unopt-B	81.17	48.03	17.27	6.55
PBC		20.84	10.96	10.00	1.74
<i>Winner: PBC (EE Diff: 60.32 mm)</i>					

4 Extreme Cases Analysis

4.1 Best Performance

Configuration:

- Controller: PBC
- Uncertainty: extreme
- Initial condition: matched
- Gains: optimized
- Trajectory: bang_coast_bang (label A)

Metrics:

- EE max error: 2.40 mm
- EE RMS error: 1.99 mm
- Joint max error: 0.82 deg
- Joint RMS error: 0.40 deg

4.2 Worst Performance

Configuration:

- Controller: FBL
- Uncertainty: extreme
- Initial condition: matched
- Gains: unoptimized
- Trajectory: bang_bang (label B)

Metrics:

- EE max error: **83.23** mm
- EE RMS error: 50.97 mm
- Joint max error: 17.23 deg
- Joint RMS error: 6.65 deg

5 Direct Controller Comparisons

This section presents head-to-head comparisons between FBL and PBC controllers under identical conditions, using the End-Effector Max Error [mm] metric. The better value is **Green**. The configuration is abbreviated to *Profile/Unc/Match-Opt/traj*.

Table 9: Head-to-head controller comparison (lower is better)

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
BB /extreme/Match-Opt (A)	19.05	2.41	16.64
BB /extreme/Match-Opt (B)	31.87	11.67	20.20
BCB /extreme/Match-Opt (A)	19.02	2.40	16.62
BCB /extreme/Match-Opt (B)	31.64	11.74	19.90
Quintic /extreme/Match-Opt (A)	19.10	2.41	16.69
Quintic /extreme/Match-Opt (B)	37.83	11.63	26.20
BB /extreme/Match-Unopt (A)	39.27	2.88	36.39
BB /extreme/Match-Unopt (B)	83.23	14.60	68.63
BCB /extreme/Match-Unopt (A)	39.23	2.88	36.35
BCB /extreme/Match-Unopt (B)	82.70	14.55	68.15
Quintic /extreme/Match-Unopt (A)	39.80	2.89	36.92

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
Quintic /extreme/Match-Unopt (B)	81.15	14.56	66.59
BB /extreme/Mismatch-Opt (A)	20.84	20.84	0.00
BB /extreme/Mismatch-Opt (B)	31.87	20.84	11.02
BCB /extreme/Mismatch-Opt (A)	20.84	20.84	0.00
BCB /extreme/Mismatch-Opt (B)	31.64	20.84	10.80
Quintic /extreme/Mismatch-Opt (A)	20.84	20.84	0.00
Quintic /extreme/Mismatch-Opt (B)	37.83	20.84	16.98
BB /extreme/Mismatch-Unopt (A)	39.07	20.84	18.22
BB /extreme/Mismatch-Unopt (B)	81.44	20.84	60.60
BCB /extreme/Mismatch-Unopt (A)	39.05	20.84	18.21
BCB /extreme/Mismatch-Unopt (B)	81.17	20.84	60.32
Quintic /extreme/Mismatch-Unopt (A)	39.52	20.84	18.67
Quintic /extreme/Mismatch-Unopt (B)	82.13	20.84	61.28
BB /high/Match-Opt (A)	10.17	6.10	4.07
BB /high/Match-Opt (B)	14.04	6.38	7.66
BCB /high/Match-Opt (A)	10.15	6.09	4.06
BCB /high/Match-Opt (B)	13.68	6.38	7.31
Quintic /high/Match-Opt (A)	10.19	6.10	4.09
Quintic /high/Match-Opt (B)	14.54	6.36	8.18
BB /high/Match-Unopt (A)	21.04	7.31	13.74
BB /high/Match-Unopt (B)	48.51	7.89	40.61
BCB /high/Match-Unopt (A)	21.02	7.30	13.72
BCB /high/Match-Unopt (B)	48.56	7.85	40.71
Quintic /high/Match-Unopt (A)	20.93	7.31	13.62
Quintic /high/Match-Unopt (B)	48.28	7.87	40.41
BB /high/Mismatch-Opt (A)	20.84	20.84	0.00
BB /high/Mismatch-Opt (B)	20.84	20.84	0.00
BCB /high/Mismatch-Opt (A)	20.84	20.84	0.00

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
BCB /high/Mismatch-Opt (B)	20.84	20.84	0.00
Quintic /high/Mismatch-Opt (A)	20.84	20.84	0.00
Quintic /high/Mismatch-Opt (B)	20.84	20.84	0.00
BB /high/Mismatch-Unopt (A)	20.84	20.84	0.00
BB /high/Mismatch-Unopt (B)	48.51	20.84	27.67
BCB /high/Mismatch-Unopt (A)	20.84	20.84	0.00
BCB /high/Mismatch-Unopt (B)	48.56	20.84	27.72
Quintic /high/Mismatch-Unopt (A)	20.99	20.84	0.15
Quintic /high/Mismatch-Unopt (B)	48.26	20.84	27.42

5.1 Overall Winner Statistics

- FBL wins: 11 (22.9%)
- PBC wins: 37 (77.1%)
- Total comparisons: 48